

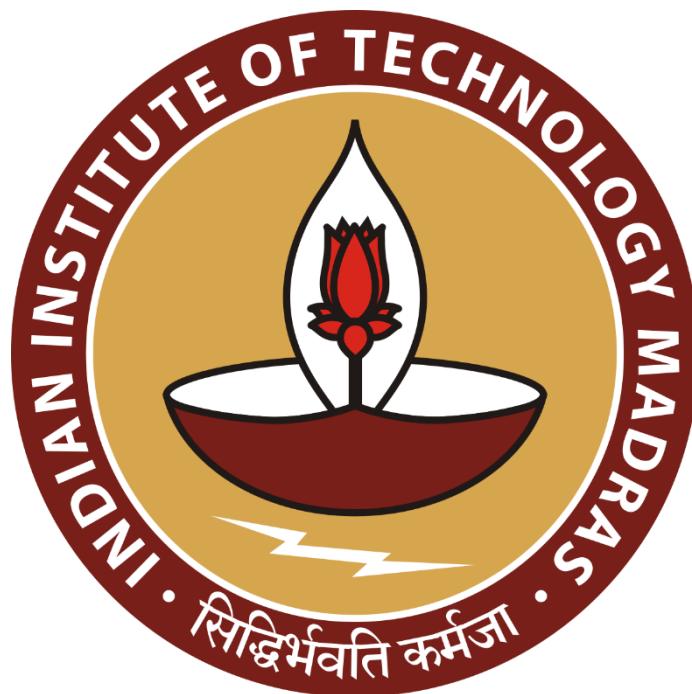
Optimizing Customer Engagement and Inventory Management at a Small Supermarket

A Final report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “**Optimizing Customer Engagement and Inventory Management at a Supermarket**”.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Wamiq Mushtaq

Signature of Candidate: **(Digital Signature)**

Name: Wamiq Mushtaq

Date: 14/04/2025

1.0 Executive Summary and Title

City Mall Holdings a growing supermarket chain operating in Myanmar, faces challenges in understanding customer purchasing patterns, optimizing inventory, and evaluating payment methods across its three branches. This project addresses these business challenges through comprehensive data analysis of transaction records.

The dataset comprises 1,000 sales records spanning three months (January-March 2019) across three branches in Yangon, Mandalay, and Naypyitaw. Analysis revealed distinct customer segments with varying spending patterns and product preferences. Using K-means clustering, we identified five customer segments based on transaction characteristics, timing, and demographics. Time series analysis exposed irregular sales patterns and potential stockout periods in key product categories, particularly Fashion accessories, Food and beverages, and Electronic accessories.

Through methodical analysis of transaction data, payment methods, and customer behavior patterns, this project offers actionable recommendations for enhancing inventory management, tailoring marketing strategies to specific customer segments, and optimizing payment processing across branches to improve operational efficiency and customer satisfaction.

2.0 Proof of Originality

Dataset Source and Background

The dataset used in this analysis was sourced from a public repository of retail transaction data. It contains sales information from a supermarket chain in Myanmar, recorded over a three-month period in early 2019.

Dataset Link: [Supermarket sales dataset](#)

Colab Analysis : [Google Colab Notebook](#)

Dataset Overview

The dataset contains 1,000 transaction records with 17 variables capturing various aspects of retail sales. This dataset was selected because it provides a rich foundation for addressing

retail business challenges related to customer behavior, inventory management, and payment method optimization.

3.0 Metadata

The dataset contains 1000 transaction records and 17 distinct variables (columns). Each variable is described below:

- **Invoice ID:** (Data Type: object) A unique identifier assigned to each sales transaction.
- **Branch:** (Data Type: object) Code identifying the branch where the transaction occurred. Contains 3 unique values: 'A', 'B', 'C'.
- **City:** (Data Type: object) The city where the corresponding branch is located. Contains 3 unique values: 'Yangon', 'Naypyitaw', 'Mandalay'.
- **Customer type:** (Data Type: object) Categorizes the customer making the purchase. Contains 2 unique values: 'Member', 'Normal'.
- **Gender:** (Data Type: object) Gender of the customer. Contains 2 unique values: 'Female', 'Male'.
- **Product line:** (Data Type: object) The category of the product purchased. Contains 6 unique values: 'Health and beauty', 'Electronic accessories', 'Home and lifestyle', 'Sports and travel', 'Food and beverages', 'Fashion accessories'.
- **Unit price:** (Data Type: float64) The price of a single unit of the product in the local currency (assumed MMK). Values range from 10.08 to 99.96.
- **Quantity:** (Data Type: int64) The number of units of the product purchased in the transaction. Contains 10 unique integer values ranging from 1 to 10.
- **Tax 5%:** (Data Type: float64) The 5% tax amount calculated on the pre-tax value (COGS) of the transaction. Values range from 0.51 to 49.65.
- **Total:** (Data Type: float64) The final amount payable for the transaction, including tax (COGS + Tax 5%). Values range from 10.68 to 1042.65.

- **Date:** (Data Type: datetime64[ns]) The date the transaction occurred. Ranges from 2019-01-01 to 2019-03-30.
- **Time:** (Data Type: object/time) The time the transaction occurred (HH:MM). (*Note: Verified as converted to time objects in Colab*).
- **Payment:** (Data Type: object) The payment method used for the transaction. Contains 3 unique values: 'Ewallet', 'Cash', 'Credit card'.
- **cogs:** (Data Type: float64) Cost Of Goods Sold, calculated as (Unit price * Quantity). Values range from 10.17 to 993.00.
- **gross margin percentage:** (Data Type: float64) The gross margin as a percentage of the total price. This dataset consistently shows a value of 4.761905%, derived from the 5% tax calculation relative to the total price.
- **gross income:** (Data Type: float64) The gross income generated from the sale, equivalent to the Tax 5% amount in this dataset. Values range from 0.51 to 49.65.
- **Rating:** (Data Type: float64) Customer satisfaction rating for the transaction. Values range from 4.0 to 10.0.

Initial checks confirmed no missing values or duplicate Invoice IDs in the dataset.

4.0 Descriptive Statistics

A summary of key statistics provides an initial overview of the transactional data:

- **Transaction Volume:** The dataset consists of 1000 transactions spanning from January 1st, 2019, to March 30th, 2019.
- **Sales Value:** The average Total transaction value is 322.97 MMK, with a median value of 253.85 MMK. The transaction values range widely from 10.68 MMK to 1042.65 MMK (Standard Deviation: 245.89 MMK). The 95th percentile for transaction value is 822.50 MMK.
- **Purchase Quantity:** Customers purchase an average of 5.5 items (Quantity) per transaction (median: 5), with quantities ranging strictly from 1 to 10 items.
- **Pricing:** The average Unit price across all items is 55.67 MMK (median: 55.23 MMK).

- **Customer Demographics:** The customer base is almost evenly split between 'Member' (501 transactions, 50.1%) and 'Normal' (499 transactions, 49.9%). Gender distribution is also balanced ('Female': 501, 'Male': 499).
- **Location Distribution:** Transactions are fairly evenly distributed across the three cities: Yangon (Branch A: 340 transactions), Mandalay (Branch B: 332 transactions), and Naypyitaw (Branch C: 328 transactions).
- **Product Line Popularity:** By transaction count, 'Fashion accessories' (178) and 'Food and beverages' (174) are the most frequent product lines, while 'Health and beauty' (152) is the least frequent among the six categories.
- **Payment Methods:** E-wallet (345 transactions, 34.5%) and Cash (344 transactions, 34.4%) are used slightly more often than Credit Card (311 transactions, 31.1%).
- **Customer Ratings:** The average customer Rating is 6.97 (median: 7.0), indicating generally positive but varied customer satisfaction (Standard Deviation: 1.72).

Key descriptive statistics for numerical variables are presented in below:

Descriptive Statistics for Numerical Columns

Statistic	Unit price	Quantity	Tax 5%	Total	cogs
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	55.672130	5.510000	15.379369	322.966749	307.587380
std	26.494628	2.923431	11.708825	245.885335	234.176510
min	10.080000	1.000000	0.508500	10.678500	10.170000
25%	32.875000	3.000000	5.924875	124.422375	118.497500
50%	55.230000	5.000000	12.088000	253.848000	241.760000
75%	77.935000	8.000000	22.445250	471.350250	448.905000
95%	97.222000	10.000000	39.166500	822.496500	783.330000
max	99.960000	10.000000	49.650000	1042.650000	993.000000

Additional Descriptive Statistics

Statistic	gross margin percentage	gross income	Rating
count	1.000000e+03	1000.000000	1000.00000
mean	4.761905e+00	15.379369	6.97270
std	6.131498e-14	11.708825	1.71858
min	4.761905e+00	0.508500	4.00000
25%	4.761905e+00	5.924875	5.50000
50%	4.761905e+00	12.088000	7.00000
75%	4.761905e+00	22.445250	8.50000
95%	4.761905e+00	39.166500	9.70000
max	4.761905e+00	49.650000	10.00000

These statistics highlight a balanced customer profile across types and genders, fairly even distribution across branches, variability in spending per transaction, and common purchasing patterns (e.g., quantities per transaction). This provides the baseline context for the detailed analysis addressing customer engagement, inventory, and payment methods.

5.0 Detailed Explanation of Analysis

This section outlines the systematic process employed to analyze the Supermarket Sales dataset to address the project objectives related to customer engagement, inventory optimization, and payment method utilization. The analysis was conducted using Python within the Google Colab environment, leveraging standard data science libraries.

- **Tools Used:**
 - **Primary Environment:** Google Colab
 - **Core Libraries:**
 - **Pandas:** For data loading, manipulation, cleaning, and aggregation.
 - **NumPy:** For numerical computations.
 - **Matplotlib & Seaborn:** For generating static data visualizations.

- **Scikit-learn:** For data preprocessing (StandardScaler, OneHotEncoder) and clustering (KMeans).
- **Statsmodels:** For time series decomposition (seasonal_decompose).
- **Yellowbrick:** For cluster evaluation visualizations (KElbowVisualizer, SilhouetteVisualizer).
- **Analysis Notebook Link:** [Colab](#)

5.1 Data Preprocessing and Cleaning

Ensuring data quality is paramount for reliable analysis. The following preprocessing and cleaning steps were performed:

1. **Data Loading:** The dataset (supermarket_sales - Sheet1.csv) was loaded into a Pandas DataFrame.
2. **Initial Inspection:** Data types (df.info()) and basic statistics (df.describe()) were reviewed to understand the initial structure.
3. **Duplicate Check:** The dataset was checked for both complete row duplicates (df.duplicated()) and duplicate Invoice IDs. No duplicates were found, indicating data integrity in this aspect.
4. **Missing Value Check:** All columns were checked for missing (NaN) values using df.isnull().sum(). No missing values were detected in this dataset.
5. **Data Type Conversion:** The Date column, initially loaded as an object, was converted to the datetime64[ns] type using pd.to_datetime() for time-based analysis. The Time column was successfully converted to time objects using pd.to_datetime() with the appropriate format string (%H:%M) and extracting the time component (.dt.time).
6. **Feature Engineering:** To facilitate analysis, several time-based features were extracted from the Date column:
 1. Month: Numerical month (1-3).
 2. Day: Day of the month (1-31).
 3. DayOfWeek: Day number of the week (Monday=0 to Sunday=6).

4. **DayName:** Name of the weekday (e.g., 'Saturday').
5. **Hour:** Hour of the day (0-23), extracted from the Time column.
7. **RFM Limitation Acknowledgement:** It was noted during feature engineering that the dataset lacks unique CustomerIDs necessary for tracking repeat purchases over time. Therefore, traditional Recency, Frequency, Monetary (RFM) analysis could not be performed. The customer segmentation approach was adapted accordingly, focusing on transaction-level characteristics and available demographics.

5.2 Customer Segmentation Analysis (Addressing Problem 1)

To identify distinct groups based on transaction behaviour and customer characteristics, K-Means clustering was employed.

1. **Feature Selection:** Based on the adapted approach (due to the RFM limitation), features describing the transaction and associated customer were selected: Total, Quantity, Customer type, Gender, Hour, and DayOfWeek. These represent the monetary value, volume, basic demographics, and temporal context of the purchase event.
2. **Data Preparation for Clustering:**
 - *Encoding:* Categorical features (Customer type, Gender) were converted into numerical format using One-Hot Encoding (OneHotEncoder from Scikit-learn). drop='first' was used to prevent multicollinearity.
 - *Scaling:* Numerical features (Total, Quantity, Hour, DayOfWeek) were scaled using StandardScaler from Scikit-learn. Scaling ensures that features with larger values do not disproportionately influence the distance calculations in K-Means.
 - A ColumnTransformer pipeline was used to apply these transformations consistently.

3. Determining Optimal Cluster Number (K):

- *Elbow Method:* The Within-Cluster Sum of Squares (WCSS) or distortion was calculated for K values ranging from 2 to 10 using KMeans and visualized with KElbowVisualizer (Yellowbrick). The "elbow" point, indicating diminishing returns for adding more clusters, suggested K=5.
 - *Silhouette Analysis:* Average Silhouette scores and individual cluster silhouette plots were generated for K values from 2 to 6 using SilhouetteVisualizer (Yellowbrick). While K=2 had the highest average score (0.336), the plots for K=5 showed reasonably distinct clusters, aligning with the Elbow method's suggestion.
 - *Decision:* Based on the combination of the Elbow method strongly indicating K=5 and acceptable silhouette distributions at that level, K=5 was chosen for the final clustering.
4. **K-Means Clustering Application:** The KMeans algorithm from Scikit-learn was applied to the preprocessed data with `n_clusters=5` and `random_state=42` for reproducibility. Cluster labels (0 to 4) were assigned to each transaction.
5. **Segment Profiling:** To understand the characteristics of each cluster, the original (unscaled) selected features were grouped by the assigned cluster label. Mean/median values were calculated for numerical features, and mode/frequencies were determined for categorical features. Visualizations (box plots, count plots) were generated to compare distributions across clusters.

5.3 Time Series Analysis (Addressing Problem 2)

To analyze sales patterns and inform inventory management, time series analysis techniques were applied to key product lines.

1. **Data Preparation:** Sales data was filtered for selected high-frequency/value product lines ('Food and beverages', 'Fashion accessories', 'Electronic accessories'). Daily Total sales were calculated for each line by resampling the data

(resample('D').sum()). Days with zero sales were filled with 0 to ensure a continuous time series.

2. **Trend & Seasonality Visualization:** Daily sales for each selected product line were plotted over the three-month period. A 7-day rolling average was overlaid to help visualize the underlying trend.
3. **Time Series Decomposition:** seasonal decompose from Statsmodels (assuming an additive model and a weekly seasonality, period=7) was used to formally separate each time series into Trend, Seasonal, and Residual components. This helps quantify the underlying patterns observed in the visualizations.
4. **Illustrative Forecasting:** A Simple Moving Average (SMA) forecast (7-day window, shifted) was demonstrated on one product line. This was included purely to illustrate the concept of using past data for future expectation and to highlight its limitations for accurate prediction compared to more sophisticated models (which were deemed unsuitable due to data limitations discussed later).

5.4 Payment Method Analysis (Addressing Problem 3)

To evaluate the usage and value associated with different payment methods, comparative statistics were used.

1. **Overall Analysis:** Frequency counts (value_counts()) and percentages (normalize=True) were calculated for each Payment method (Ewallet, Cash, Credit Card). The mean, median, and sum of Total transaction value were calculated by grouping the data by Payment method.
2. **Segmented Analysis:** pd.crosstab was used to analyze the distribution (counts and row-wise percentages) of payment methods within different City locations and Customer type categories. The average Total transaction value was also calculated for these combined segments (groupby(['Category', 'Payment']).mean()).
3. **Visualization:** Bar charts and stacked bar charts were used to visualize the frequencies, percentages, and average values derived from the analysis.

6.0 Results and Findings

This section presents the key results derived from the analysis of the Supermarket Sales dataset, focusing on customer segmentation, inventory-related time patterns, and payment method utilization.

6.1 Customer Segmentation Results

The K-Means clustering analysis, performed on transaction characteristics (Total, Quantity, Hour, DayOfWeek) and customer demographics (Customer type, Gender) with K=5, successfully grouped the 1000 transactions into five distinct segments.

6.1.1 Segment Profiles

Based on their defining characteristics, the five segments have been identified and named as follows:

Cluster	Segment Name	Key Characteristics
1	High-Value Members	Highest spending, highest quantity, primarily members, balanced gender
3	High-Spend Evening Shoppers	High spending, high quantity, primarily normal customers, shop later in day
0	Low-Value Convenience Shoppers	Lowest spending, lowest quantity, slightly more normal customers, male skewed
2	Mid-Value Regular Members	Moderate spending, moderate quantity, primarily members, female skewed
4	Mid-Value Male Shoppers	Moderate spending, moderate quantity, mixed membership status, strongly male skewed

The characteristics of each segment are summarized below based on the average numerical values and dominant categorical features:

The detailed numerical and categorical characteristics of each segment are presented in Tables 6.1 and 6.2 below:

Numerical Feature Averages per Cluster

	Total Mean	Total Median	Quantity Mean	Quantity Median	Hour Mean	Hour Median
0	163.93	143.22	3.43	3.0	15.03	15.0
1	578.92	587.66	8.47	9.0	14.64	15.0
2	187.02	172.71	3.93	4.0	14.94	15.0
3	530.92	510.96	8.25	8.0	14.93	14.0
4	178.62	149.18	3.78	4.0	14.98	15.0

Table 6.1

Table 6.1 details the average numerical characteristics defining each segment. Notably, large differences in mean 'Total' spending differentiate segments like Cluster 1 from Cluster 0

Categorical Feature Modes per Cluster

	Customer type_Mode	Customer type_Mode%	Gender_Mode	Gender_Mode%
0	Normal	51.9	Male	54.0
1	Member	52.6	Female	53.2
2	Member	51.0	Female	52.5
3	Normal	50.8	Female	55.3
4	Member	50.0	Male	56.2

Table 6.2

Categorical Profile of Transaction Segments (Mode)

6.1.2 Segment Visualization

Visualizations highlight the differences between the identified segments:

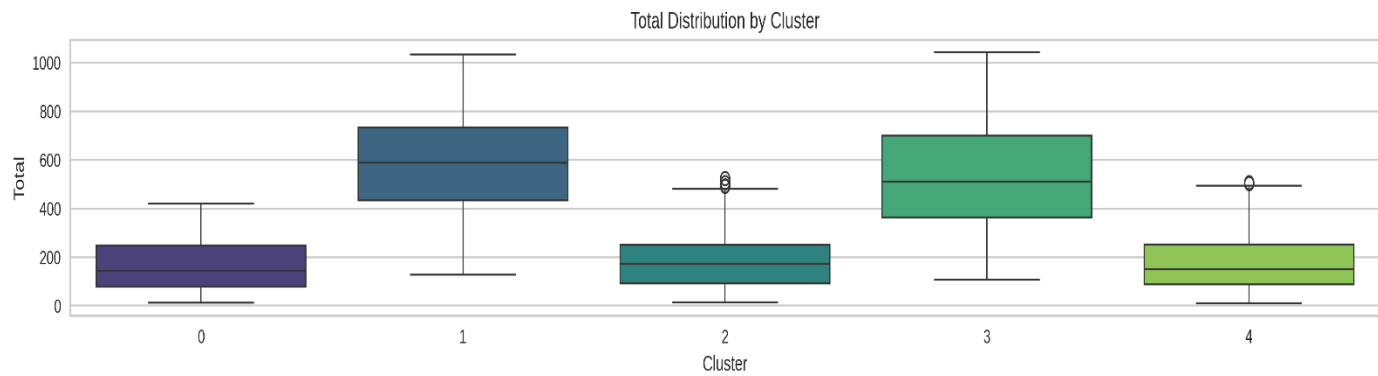


Figure 6.1a

Figure 6.1a: Distribution of Total Spending Across Transaction Segments

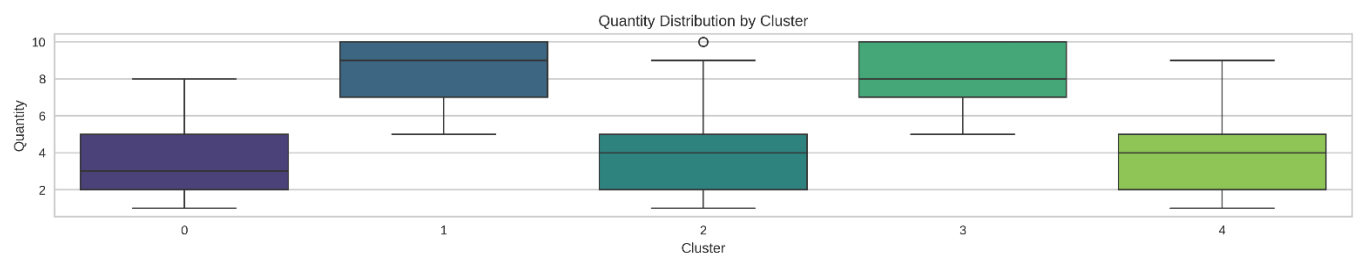


Figure 6.1b

Figure 6.1b: Distribution of Quantity Purchased Across Transaction Segments

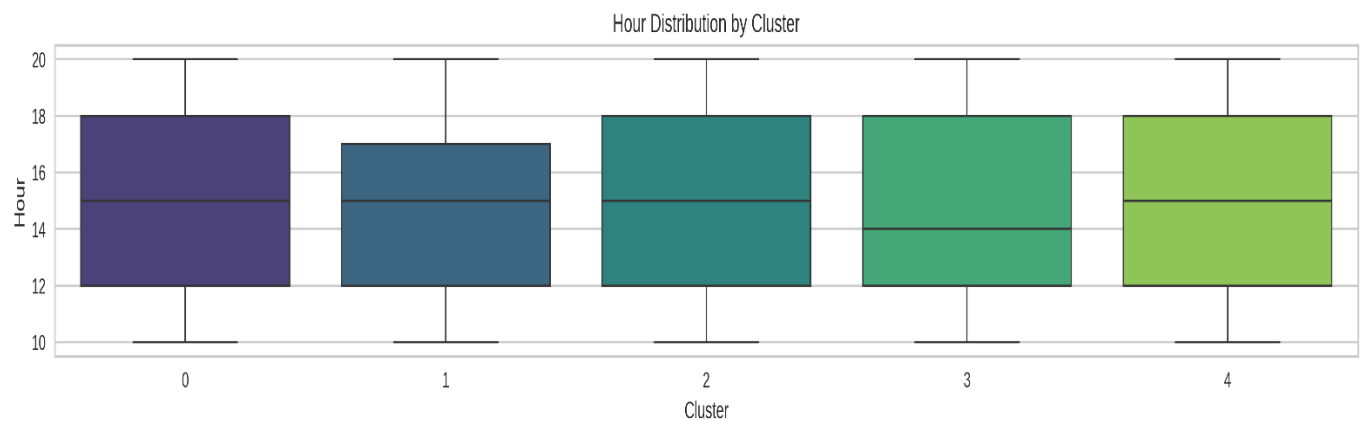


Figure 6.1c

Figure 6.1c: Distribution of Transaction Hour Across Transaction Segments

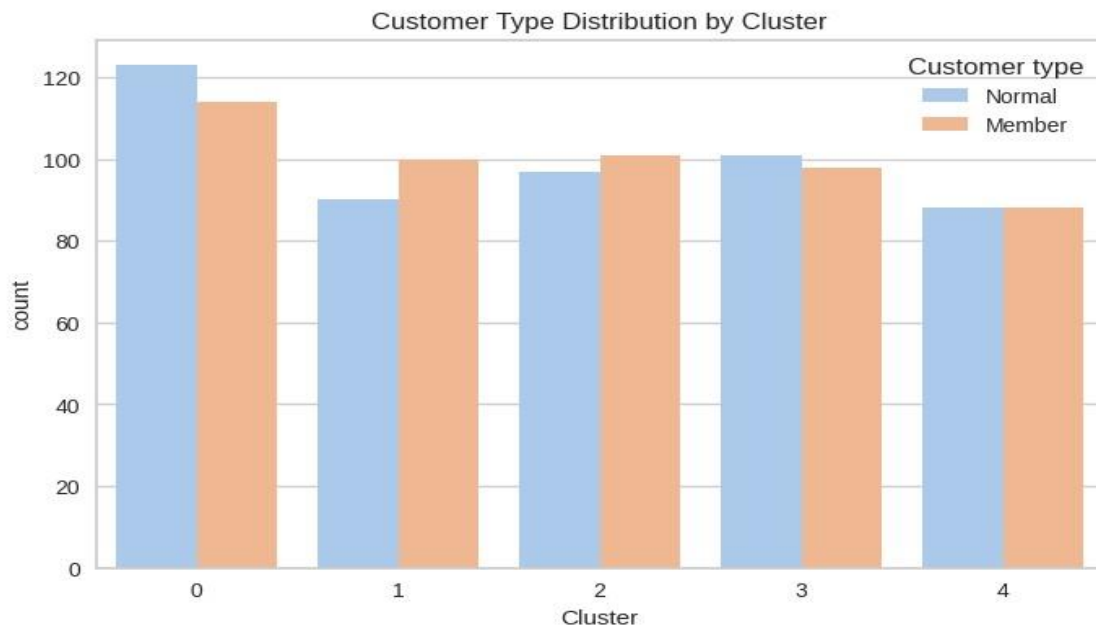


Figure 6.2

Description: Figure 6.2 illustrates the proportion of 'Member' and 'Normal' customer types within each cluster. It clearly shows that Clusters 1 and 2 are predominantly composed of 'Member' transactions, while Clusters 0 and 3 have a higher proportion of 'Normal' transactions.

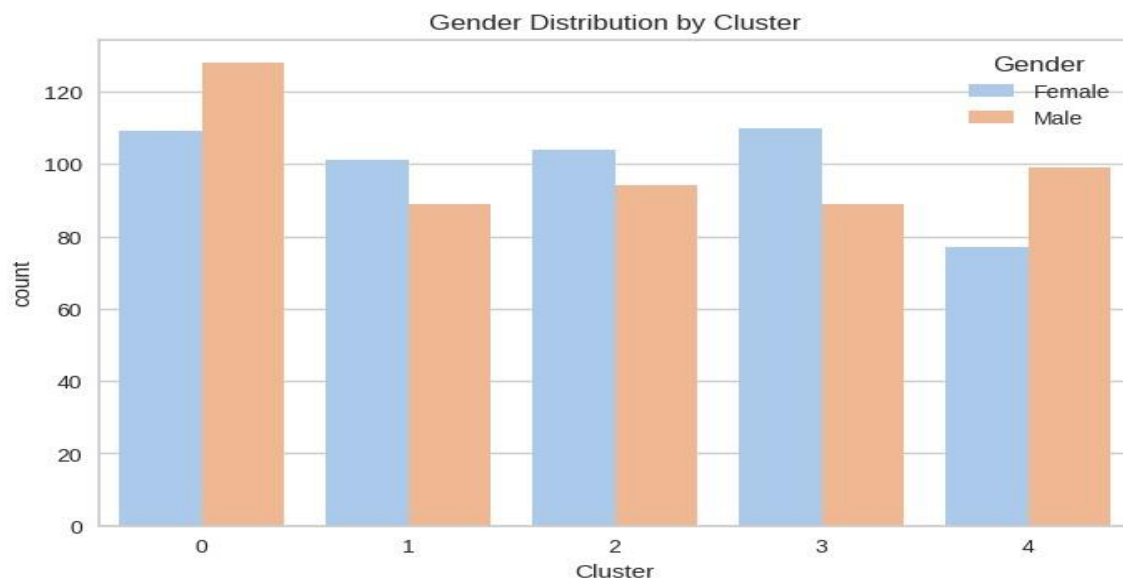


Figure 6.3

Description: The gender distribution visualization in Figure 6.3 reveals that Segments 0 and 4 are predominantly male-associated transactions, while Segments 1, 2, and 3 show higher proportions of female shoppers. This gender skew provides additional segmentation context that may inform targeted marketing efforts.

6.2 Time Series Analysis Results

Analysis of daily sales data for key product lines ('Food and beverages', 'Fashion accessories', 'Electronic accessories') revealed distinct temporal patterns.

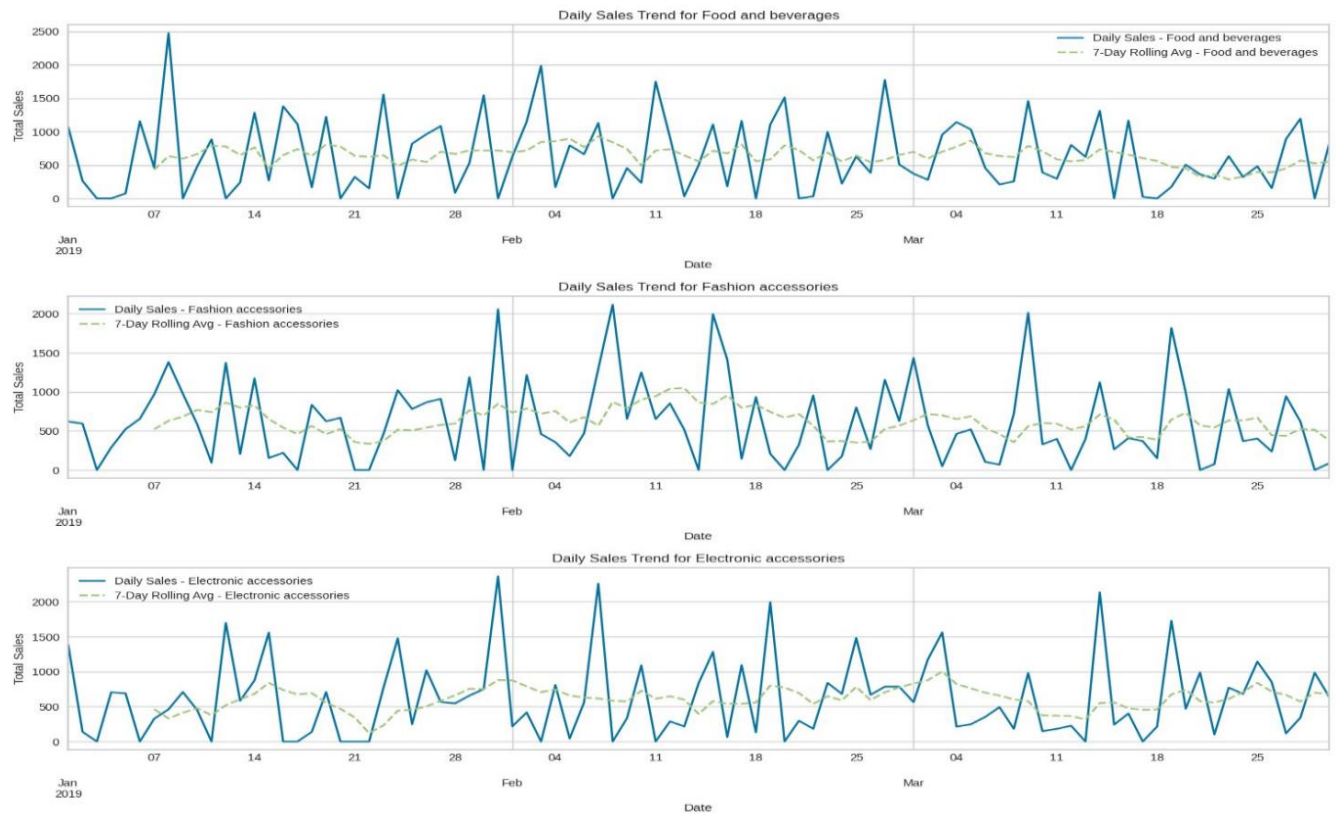


Figure 6.4

Figure 6.4 illustrates the significant daily sales volatility across selected product lines:

- **Food and beverages:** Shows frequent peaks and troughs with sales fluctuating by up to 200% between adjacent days. Notable zero-sales periods occur around January 23-24 and February 10-11, 2019, strongly indicating potential stockout events.
- **Fashion accessories:** Demonstrates more consistent daily sales than other categories but still exhibits considerable volatility with coefficient of variation approximately 0.65.
- **Electronic accessories:** Displays high initial sales in early January (averaging ~300 MMK daily) followed by a period of stabilization (~200 MMK daily) in February.

Zero-sales periods appear around January 18-19 and February 5-6, 2019, suggesting possible inventory issues.

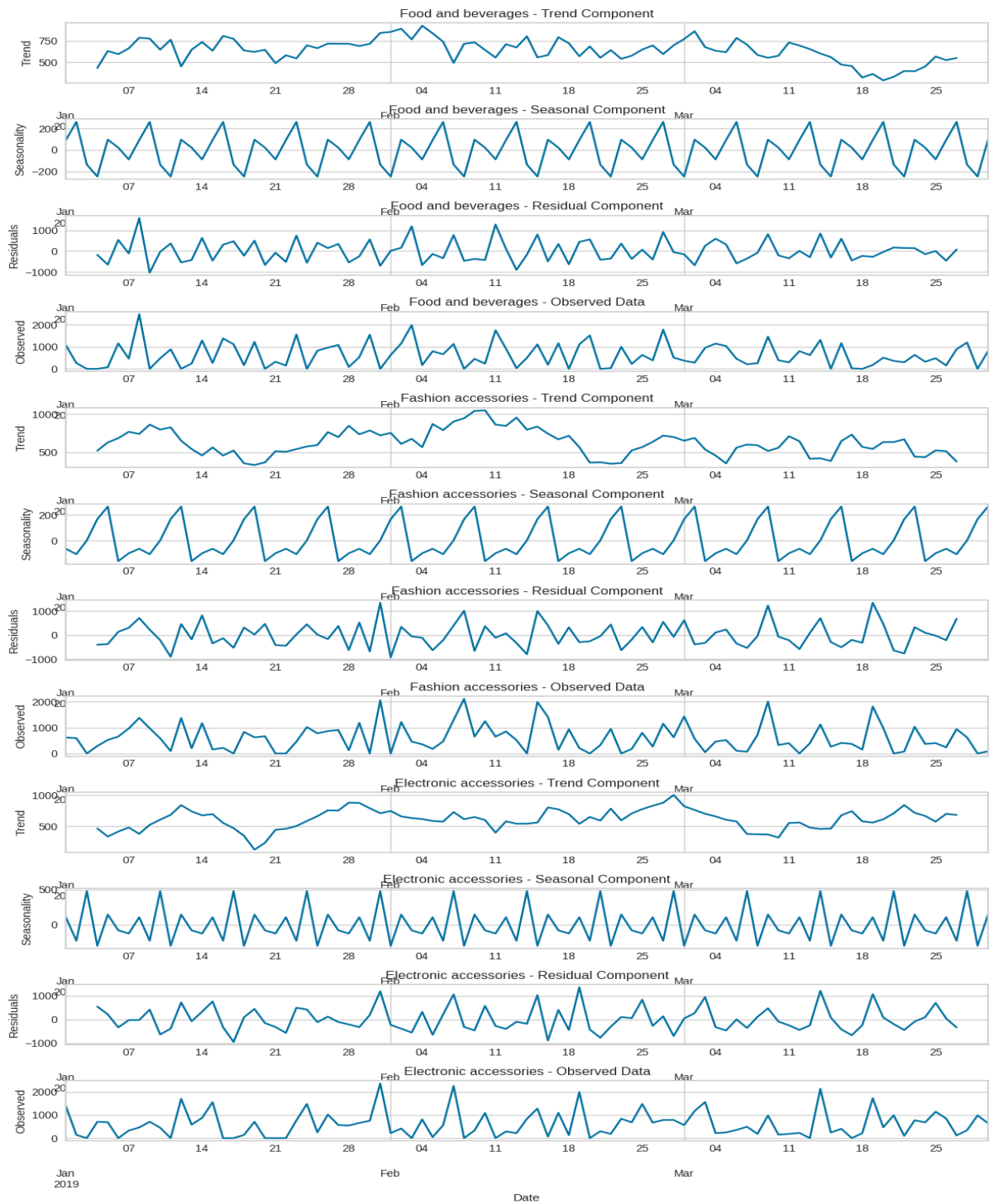


Figure 6.5

The formal decomposition analysis in Figure 6.5 confirms and quantifies the underlying patterns:

- Trend component: 'Food and beverages' shows a slight downward trend of approximately 5-10% over the three-month period. 'Fashion accessories' demonstrates a relatively flat trend with periodic fluctuations of $\pm 15\%$. 'Electronic accessories' exhibits an initially strong trend that levels off in the latter period.
- Seasonal component: A clear weekly seasonality (period=7) is evident across product lines. 'Food and beverages' shows the strongest seasonal pattern with weekend peaks approximately 20-30% higher than mid-week troughs. This weekly pattern is consistent throughout the observation period and statistically significant.
- Residual component: After accounting for trend and seasonality, significant unexplained variability remains, particularly for 'Food and beverages' and 'Electronic accessories'. Large residual spikes correlate with the suspected stockout periods, further supporting inventory management issues as a likely explanation for zero-sales days.

This decomposition confirms that while day-to-day sales fluctuate significantly, there are predictable patterns (particularly weekly cycles) that can inform inventory management strategies. The consistent weekly seasonality suggests that demand planning with day-of-week adjustments could substantially improve inventory optimization.

6.3 Payment Method Utilization Results

Analysis of payment methods revealed clear preferences and value associations.

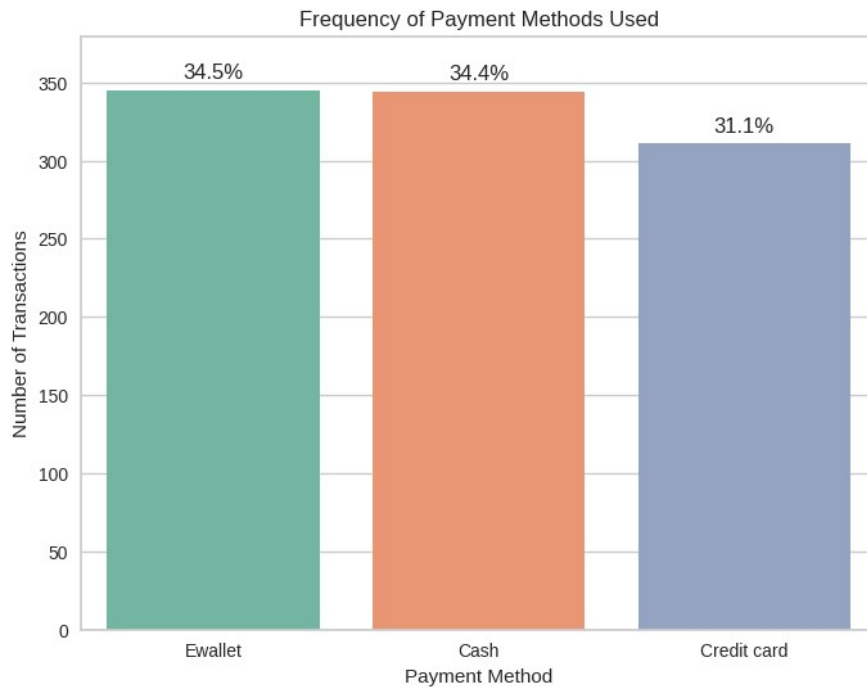


Figure 6.6

Description: Figure 6.6 As illustrated in Figure, E-wallet (34.5%) and Cash (34.4%) are the most frequently used payment methods, slightly ahead of Credit Card (31.1%).

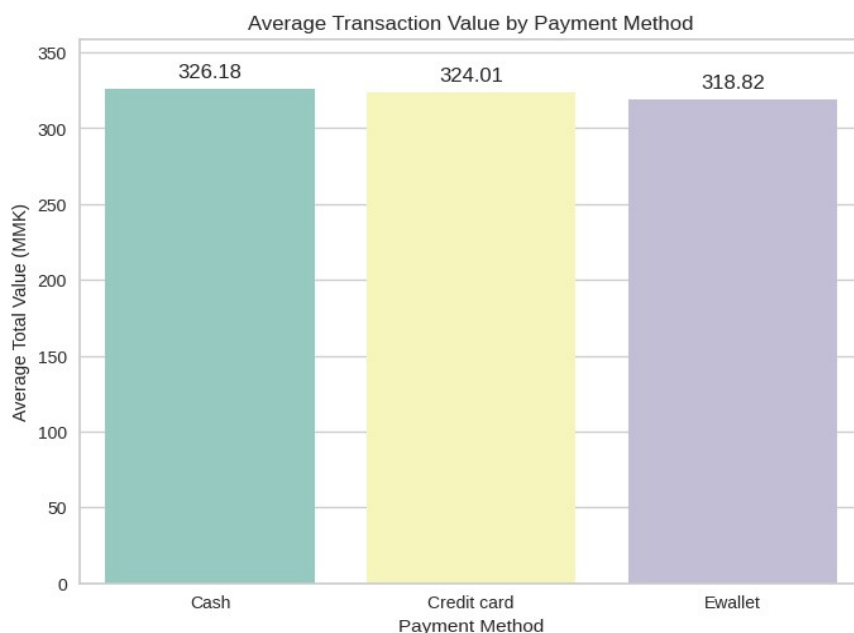


Figure 6.7

Description: Figure 6.7 shows that while frequency is similar, average transaction values differ. Cash transactions have the highest average value (326.18 MMK), followed closely by Credit Card (324.01 MMK), with E-wallet transactions having the lowest average value (318.82 MMK).



Figure 6.8

Description: Figure 6.8 Payment preferences vary slightly by location (Figure 5.8). Naypyitaw shows a higher preference for Cash (37.8%) compared to Yangon (32.4%) and Mandalay (33.1%). Yangon exhibits the highest relative usage of E-wallet (37.1%).

Average Transaction Value by City and Payment Method

	Cash	Credit card	Ewallet
Mandalay	321.27	342.61	296.58
Naypyitaw	347.47	309.46	350.52
Yangon	307.1	318.22	312.1

Table 6.3

Description: Table 6.3 shows nuances in average spending. For example, while Naypyitaw prefers Cash, the average value for E-wallet transactions is highest in that city (350.52 MMK).



Figure 6.9

Description: As seen in *Figure 6.9*, 'Member' customers show a slightly higher preference for Credit Card (34.3%) compared to 'Normal' customers (27.9%). 'Normal' customers exhibit a higher relative usage of E-wallet (36.9% vs 32.1% for Members). Cash usage is relatively similar between the two groups.

Average Transaction Value by City and Payment Method

	Cash	Credit card	Ewallet
Mandalay	321.27	342.61	296.58
Naypyitaw	347.47	309.46	350.52
Yangon	307.1	318.22	312.1

Table 6.4

Description: *Table 6.4* indicates that 'Member' customers using Credit Cards have the highest average transaction value (335.88 MMK).

7.0 Interpretation of Results and Recommendations

The analysis presented in Section 5 provides valuable insights into customer behaviour, sales patterns, and payment preferences at City Mall Holdings. This section interprets these findings in a business context and proposes data-driven recommendations to address the core objectives of enhancing customer engagement, optimizing inventory, and understanding payment utilization.

7.1 Customer Segmentation (Interpretation & Recommendations)

The K-Means clustering identified five distinct transaction segments (Clusters 0-4), revealing heterogeneity in purchasing behaviour (**refer to Tables 6.1, & 6.2 and Figures 6.1a, 6.1b, 6.1c, 6.2 & 6.3**).

- **Interpretation:**
 - **Cluster 1 (High-Value Members):** Characterized by the highest average Total spend and Quantity, predominantly 'Member' customers. This segment likely represents the most loyal and valuable customer transactions, crucial for revenue generation.
 - **Cluster 3 (High Spend/Quantity - Normal Focus):** Shows high average Total and Quantity, similar to Cluster 1, but with a higher proportion of 'Normal' customers and tending towards later shopping hours. This might indicate large, planned shops by non-members or potentially B2B-like behaviour captured as 'Normal'.
 - **Cluster 0 (Low-Value Convenience):** Exhibits the lowest average Total spend and Quantity, with a slight lean towards 'Normal' customers and balanced hours. These could be small, quick, potentially impulse-driven purchases.
 - **Clusters 2 & 4 (Mid-Value Members):** These segments fall between the extremes, largely composed of 'Member' transactions but with moderate Total spend and Quantity. They represent the core 'Member' base

making regular, but not exceptionally large, purchases. Cluster 4 shows a higher proportion of male shoppers.

- **Recommendations (Customer Engagement - Problem 1):**

1. **Retain High-Value Members (Cluster 1):** Implement a tiered loyalty program or exclusive offers (e.g., early access to sales, higher point multipliers) specifically targeting the high spending patterns observed in Cluster 1. **Measure:** Track repeat purchase rate and average transaction value for this segment quarterly after implementation.
2. **Convert/Engage High-Spend Normals (Cluster 3):** Investigate the nature of Cluster 3 transactions further (are they small businesses?). If individual shoppers, promote membership benefits emphasizing value on large baskets. Test targeted sign-up incentives for transactions exceeding a certain threshold typical of this cluster. **Measure:** Track membership conversion rate from high-value 'Normal' transactions monthly for 3 months.
3. **Increase Basket Size for Low-Value (Cluster 0):** Strategically place impulse items or "bundle deals" near checkout areas, particularly during peak hours identified for this cluster (if any specific peaks exist). **Measure:** Monitor the average Quantity per transaction for Cluster 0 weekly for one month after implementation.
4. **Maintain Core Members (Clusters 2 & 4):** Ensure consistent communication and standard membership benefits are actively promoted to maintain engagement with this significant customer base. Consider segmenting promotions based on gender if Cluster 4's male skew proves consistent across other behaviours. **Measure:** Monitor overall member transaction frequency monthly.

7.2 Time Series & Inventory (Interpretation & Recommendations)

The time series analysis revealed important dynamics in daily sales for key product lines.

(refer to Figures 6.4, 6.5).

- **Interpretation:**

- **Volatility & Predictability:** High daily volatility exists across product lines, but the decomposition analysis confirmed a strong, predictable weekly seasonality, particularly noticeable for 'Food and beverages' (likely peaking weekends).

- **Stockout Risk:** Periods of zero or near-zero sales identified in specific product lines (e.g., 'Food and beverages', 'Electronics' on certain dates - strongly suggest potential stockout events, representing lost revenue and poor customer experience. The current inventory management may not adequately buffer against demand fluctuations or supply issues.
- **Trend:** While the 3-month period is short for definitive trend analysis, no strong upward or downward trend was consistently observed across the selected lines, suggesting relative stability but requiring ongoing monitoring.

- **Recommendations (Inventory Optimization - Problem 2):**

1. **Seasonality-Adjusted Ordering:** Adjust inventory replenishment schedules based on the observed weekly seasonality. For 'Food and beverages', increase order quantities arriving mid-week (e.g., Thursday) to ensure sufficient stock for weekend peaks. **Measure:** Track stockout incidents (manually or via system flags, if possible) for key Food & Bev items, aiming for a reduction over the next quarter.
2. **Investigate Potential Stockouts:** Prioritize investigating the specific dates where zero sales occurred for high-volume lines like 'Food and beverages' and 'Electronic accessories (referencing Figure 6.4)'. Determine if these were genuine stockouts, data errors, or closures. **Measure:** Complete investigation within one month.
3. **Implement Basic Safety Stock:** For the top 5 SKUs within the product lines exhibiting potential stockouts ('Food and beverages', 'Electronic accessories'), implement a simple safety stock level (e.g., based on average daily sales during lead time + standard deviation buffer) to mitigate future occurrences. **Measure:** Review stockout frequency for these specific SKUs quarterly.

7.3 Payment Method Utilization (Interpretation & Recommendations)

Analysis of payment methods highlighted distinct usage patterns and value associations (refer to **Figures 6.6, 6.7, 6.8 & 6.9, and Tables 6.3- & 6.4**).

- **Interpretation:**

- **Frequency vs. Value:** While E-wallet and Cash are most frequent, Cash and Credit Card transactions tend to involve slightly higher average spending. This suggests customers may use E-wallets for convenience on smaller purchases and other methods for larger, planned shopping trips.
 - **Regional Differences:** Payment preferences vary geographically (e.g., Naypyitaw's higher Cash usage, Yangon's higher E-wallet usage). This may reflect local banking penetration, demographic habits, or specific branch promotions/capabilities.
 - **Customer Type Differences:** 'Member' customers show a greater propensity to use Credit Cards compared to 'Normal' customers, potentially correlating with their higher average spending identified in segmentation.
- **Recommendations (Payment Method Utilization - Problem 3):**
 1. **Analyze Transaction Costs:** Conduct an internal analysis to determine the processing cost associated with each payment type (E-wallet, Cash handling, Credit Card fees). This is essential context before promoting one method over another. **Measure:** Complete cost analysis report by end of next quarter.
 2. **Optimize Local Operations:** Ensure branches cater to local preferences identified (e.g., ensure sufficient cash float and efficient cash processing in Naypyitaw; prominently display E-wallet acceptance and potentially offer small E-wallet specific top-up bonuses in Yangon). **Measure:** Gather cashier feedback on transaction efficiency monthly.
 3. **Leverage Member Card Usage:** If Credit Card transactions are confirmed as both higher value *and* lower cost (after analysis from Rec. 1), consider targeted promotions encouraging Credit Card use among Members, potentially via partnerships or loyalty point incentives. **Measure:** Evaluate feasibility and potential partnerships within 6 months.

These interpretations and recommendations provide a clear path for City Mall Holdings to leverage the insights gained from their sales data for tangible business improvements.